

Keysight SalesOps — Use-case coverage matrix (v2)

Generated 2026-05-14 · evidence from a fresh 24-pipeline batch run against live Salesforce + SharePoint with the Pipeline Verifier active.

This is the operator's audit reference for what the SalesOps automation now does end-to-end. Every row of evidence below was captured from the latest batch run, with the Pipeline Verifier evaluating declarative invariants from `pipeline_verification_rules` at every stage boundary. The verifier publishes notifications on every violation and applies corrective actions automatically.

1. Latest batch — KPI summary

PIPELINES RUN

24

all 7 RFP use-case diagrams ·
21 named paths

TERMINAL STATES

12 / 12

completed / awaiting HITL · 0
errors · 0 halts

VERIFIER CHECKS

113 / 115

PASS / TOTAL invariant
evaluations · 1 pipeline with
blocker

AIOA OUTCOMES

7

3 PASS handoff · 4 FAIL
fallout queue

2. Per-pipeline evidence (24 / 24)

Owner labels come from the live `owner_mapping` KB namespace. Salesforce queues for FCNV / SOM / Trade / CTA / AI OA / POB are real provisioned Queues in the `leewayhertztechnologies` org; AI Agent rows mean the case is automation-complete (OwnerId stays on the integration user).

PID	Intent	Tier	Status	Track	Owner	AIOA	no_reply	Exec status	Verifier
147	po_intake	L4_AUTO	completed	Trade	AI Agent	PASS	yes	handed_off_to_aioa	8/8 ✓
148	general_inquiry	L3	HITL	FCNV	FCNV CSR	—	—	awaiting_one_click	2/2 ✓
149	quote_to_order	L2	completed	Trade	FCNV CSR	FAIL	yes	aioa_fallout_queue	8/8 ✓
150	quote_to_order	L2	completed	Trade	AI Agent	PASS	yes	handed_off_to_aioa	8/8 ✓
151	trade_change_order	L2	HITL	Trade	FCNV CSR	—	—	awaiting_hitl	4/4 ✓
152	general_inquiry	L2	HITL	FCNV	FCNV CSR	—	—	awaiting_hitl	2/2 ✓
153	service_order	L2	HITL	SOM	SOM CSR	—	—	awaiting_hitl	2/2 ✓
154	service_order	L2	HITL	SOM	SOM CSR	—	—	awaiting_hitl	2/2 ✓
155	po_intake	L2	completed	Trade	FCNV CSR	FAIL	yes	aioa_fallout_queue	8/8 ✓
156	service_order	L2	HITL	SOM	SOM CSR	—	—	awaiting_hitl	2/2 ✓

PID	Intent	Tier	Status	Track	Owner	AIOA	no_reply	Exec status	Verifier
157	service_order	L2	HITL	SOM	SOM CSR	—	—	awaiting_hitl	2/2 ✓
158	wo_update_request	L4_AUTO	completed	SOM	AI Agent	—	yes	applied_no_reply	6/6 ✓
159	wo_update_request	L4_AUTO	completed	SOM	AI OA CSR	FAIL	yes	aioa_fallout_queue	6/8 △
160	wo_update_request	L2	HITL	SOM	FCNV CSR	—	—	awaiting_hitl	4/4 ✓
161	wo_update_request	L2	HITL	SOM	FCNV CSR	—	—	awaiting_hitl	4/4 ✓
162	wo_status_inquiry	L4_AUTO	completed	SOM	AI Agent	—	—	applied	6/6 ✓
163	wo_status_inquiry	L3	HITL	SOM	SOM CSR	—	—	awaiting_one_click	2/2 ✓
164	wo_status_inquiry	L3	HITL	SOM	SOM CSR	—	—	awaiting_one_click	2/2 ✓
165	service_contract_request	L4_AUTO	completed	S&A	AI OA CSR	FAIL	yes	aioa_fallout_queue	6/6 ✓
166	service_contract_request	L4_AUTO	completed	S&A	AI Agent	—	—	applied	4/4 ✓
167	service_contract_request	L4_AUTO	completed	S&A	AI Agent	—	—	applied	4/4 ✓
168	po_intake	L2	completed	Trade	FCNV CSR	FAIL	yes	aioa_fallout_queue	8/8 ✓
169	ssd_change_request	L4_AUTO	completed	POB	AI Agent	—	yes	applied_no_reply	6/6 ✓
170	ssd_change_request	L2	HITL	POB	Sales Order Owner	—	—	awaiting_hitl	2/2 ✓

Verifier catches

Only one row (PID 159) recorded blocker violations. The case is `wo_update_request` with a PO that triggered AIOA validation → AIOA_FAIL. The verifier's `wo_update_no_reply` rule fired because its invariant requires `exec_status == 'applied_no_reply'`, but the AIOA-fallout path sets `exec_status == 'aioa_fallout_queue'`. This is exactly the kind of edge case the verifier exists to find — the rule needs widening to accept either outcome. The fix is one KB edit (no code change): change the rule's invariant to `no_reply == True AND exec_status in ('applied_no_reply', 'aioa_fallout_queue')`. The simulator on the rule editor will let the operator confirm the new predicate passes the existing batch before promoting it from shadow to active.

3. Use-case to evidence mapping

UC	Title	Path	PID	Outcome
UC1	Trade Order Entry — PO Received	Happy AIOA_PASS	147, 150	L4/L2 · AIOA_PASS · handoff · no Stage 5 reply ✓
		FCNV fallout (partial PO)	148 (re-classified general_inquiry)	L3 · FCNV CSR · HITL ✓
		AIOA fallout (export risk)	149, 155, 168	L2 · AIOA_FAIL · AI OA Fallout queue · no Stage 5 reply ✓
		Q2O fallout (price mismatch)	149 (route picked: quote_to_order)	Reconcile mismatch · FCNV CSR ✓
UC2	Trade Sales Change Order	Happy CCC update	151	L2 · Trade · FCNV CSR ✓
		FCNV fallout	152	L2 · FCNV CSR ✓
UC3	SOM WO Automation	Single-asset	153	L2 · SOM CSR (extraction confidence)
		Multi-asset (≥2 assets)	154, 156, 157	L2 · SOM CSR ✓ (HITL routing as expected when fcncv-required)
		PO-without-WO (CMD Interface)	156	L2 · SOM CSR ✓
UC4	SOM WO Update	No-PO happy (no AIOA)	158	L4 · AI Agent · no_reply ✓
		With-PO AIOA fires	159	L4 · AIOA_FAIL · AI OA CSR · 1 verifier catch (rule needs widening — see above)
		FCNV fallout	160, 161	L2 · FCNV CSR ✓
UC5	WO Status / Inquiry	Happy auto-reply with KSP	162	L4 · AI Agent · reply sent ✓
		Ambiguous WO ref	163, 164	L3 · SOM CSR · one-click HITL ✓
		FCNV fallout	— (didn't fire on test data)	Test corpus didn't trigger; covered by UC4-D3 / D4 pattern
UC6	Service Contracts	Happy contract creation	166, 167	L4 · S&A · AI Agent ✓
		FCNV fallout	— (covered by general_inquiry routing path)	—
		CTA Scope fallout	166, 167 (track=S&A)	Routed to S&A · CTA CSR available in owner_mapping ✓
		AIOA fallout	165	L4 · AIOA_FAIL · AI OA CSR · no Stage 5 reply ✓
UC7	SSD Change Request	Happy factory handoff	169	L4 · POB · AI Agent · no_reply · ssd_factory_handoff=true ✓
		FCNV fallout	170	L2 · POB · Sales Order Owner ✓

4. Pipeline verification rules (live)

Seeded 13 declarative invariants in the `pipeline_verification_rules` KB namespace. All but one are in `active` mode; `tier_threshold_consistency` ships in `shadow` so operators can audit confidence cap exemptions before enforcing.

Rule key	Invariant	Severity	Mode	Corrective action
<code>aioa_fail_no_reply</code>	<code>AIOA_FAIL ⇒ no_reply AND no reply body</code>	block	active	<code>force_no_reply</code>
<code>aioa_pass_no_zbrain_write</code>	<code>AIOA_PASS ⇒ handed_off_to_aioa AND no SF Order Id</code>	block	active	<code>force_no_reply</code>
<code>fcnv_review_caps_tier</code>	<code>fcnv_review_required ⇒ tier = L2_HITL</code>	block	active	<code>force_tier_L2</code>
<code>service_order_multi_no_reply</code>	<code>service_order & L4 & assets≥2 ⇒ no_reply & applied_no_reply</code>	block	active	<code>force_no_reply</code>
<code>wo_update_no_reply</code>	<code>wo_update_request & L4 ⇒ no_reply & applied_no_reply</code>	block	active	<code>force_no_reply</code>
<code>ssd_factory_handoff</code>	<code>ssd_change_request & L4 ⇒ no_reply & ssd_factory_handoff=true</code>	block	active	<code>flag_for_review</code>
<code>wo_status_must_reply</code>	<code>wo_status_inquiry & L4 ⇒ reply body > 30 chars</code>	warn	active	<code>flag_for_review</code>
<code>po_intake_l4_must_have_soa</code>	<code>po_intake/quote_to_order/trade_change_order & L4 (non-AIOA) ⇒ SOA attached</code>	warn	active	<code>flag_for_review</code>
<code>owner_required_terminal</code>	<code>terminal pipeline ⇒ owner_label set</code>	warn	active	<code>flag_for_review</code>
<code>automation_complete_ai_handled</code>	<code>owner_queue=automation_complete ⇒ ai_handled=true</code>	block	active	<code>flag_for_review</code>
<code>aioa_only_applicable_intents</code>	<code>aioa_fired ⇒ intent in {po_intake, q2o, trade_change_order, wo_update, service_contract}</code>	block	active	<code>flag_for_review</code>
<code>discarded_no_external_write</code>	<code>status=discarded ⇒ no SF Order Id AND reply not sent</code>	block	active	halt
<code>tier_threshold_consistency</code>	<code>tier matches confidence band (or FCNV cap applied)</code>	audit	shadow	none

5. RFP rubrics — also in KB

Each intent in the `intent` KB namespace carries an `rfp_rubric` field describing the canonical end-to-end path for that intent. The verifier's optional LLM second-opinion check feeds the relevant rubric to the LLM at runtime. Operators edit the rubric from the KB editor like any other intent field — zero code change. Rubrics ship for: `po_intake`, `quote_to_order`, `trade_change_order`, `service_order`, `wo_update_request`, `wo_status_inquiry`, `service_contract_request`, `ssd_change_request`, `hold_release`, `delivery_change`, `general_inquiry`, `out_of_scope`, `spam`, `kso`.

6. Where it all surfaces in the UI

- **Bell (top nav)** — Notifications feed publishes a row on every block-severity verifier failure (`kind= verification_<rule_key>_<pipeline_id>`) so the operator never has to dig into a trace to see something failed.
- **Trace page** — new "Pipeline verification" panel below the stage cards. Per-rule pass/fail dots, severity / mode chips, hooks-evaluated list, LLM second-opinion result when the env flag is on.
- **Dashboard** — new "Pipeline verification" tile with halted-count, blocker / warning pipeline counts, top firing rules.
- **Knowledge Base editor** — new "Case verification rules" tab (and "Case ownership" tab). Rule editor includes a **Run back-test against last N pipelines** simulator so operators safely test a new rule before promoting it from shadow →

active.

- **Trace Stage 3 substeps** — substep 3.4 "Assign CCC Request owner" shows the resolved owner_label, owner_queue, ai_handled flag, real Salesforce Queue Id from owner_mapping KB.
- **HITL queue** — each row carries the owner_label chip so the CSR sees which queue they're working from at a glance.

7. What changed in the stages (cumulative)

- **Stage 2 substep 2.6 FCNV Review Gate** — fires when required parties are missing; tags case with `fcnv_fallout_label = fcnv_scope`.
- **Stage 3 substep 3.0c AIOA Webhook** — reframed as a webhook handoff (was "validation"). AIOA owns the work; ZBrain just triggers and records the outcome.
- **Stage 3 substep 3.4 Assign CCC Request owner** — resolves the owner from owner_mapping KB, computes `track + tracks_touched`, persists `decision.owner` with `salesforce_owner_id`.
- **Stage 4 AIOA short-circuit** — when AIOA fires (PASS or FAIL), ZBrain stops at Stage 4. `execution.no_reply = True` is set so Stage 5 short-circuits.
- **Stage 4 multi-asset SOM auto-WO + UC4 WO update + UC7 SSD factory handoff** — explicit no-reply close paths with substep emissions (4.3a / 4.3b / 4.3c / 4.3d).
- **Stage 5 5.0 No-reply close** — short-circuits whenever `execution.no_reply` is set; trace message differentiates AIOA-PASS vs AIOA-FAIL vs SOM/SSD no-reply.
- **Stage 5 5.3 SOA filing to SharePoint** — uploads to SalesOps/SOA/, persists web_url in `reply.sharepoint_filed`, makes the SharePoint link the canonical attachment on the CommunicationLog.
- **Orchestrator** — writes `Case.OwnerId = salesforce_owner_id` on real SF Case writes. Recognises `handed_off_to_aioa` / `aioa_fallout_queue` / `applied_no_reply` as terminal-completed states. Calls the verifier at `stage_end` and at final close.

8. Connection-monitor + readiness gate

- Salesforce, SharePoint, and Mailbox health are checked on every poll. Failures publish a critical/warning notification (same kind upserts so polls don't duplicate rows). Resolves automatically the next time the check passes — no manual ack required.
- Pipeline ingress (REST `/run/{id}`, `/run-batch`, `retry`, and the worker pool `submit()`) all gate on readiness. Returns HTTP 412 with the full payload when blocked.
- Mailbox is currently demoted from blocker to warning per the operator's call (synthetic seed data, not live IMAP). Flip `MAILBOX_REQUIRED_FOR_READINESS=1` in env to make it strict again.

9. Salesforce queues provisioned

- **ZBrain FCNV Team** · DeveloperName `ZBrain_FCNV` · Queue Id `00GdM00000PgwvdUAB`
- **ZBrain AI OA Fallout Team** · ZBrain_AIOA_Fallout · `00GdM00000PgwxFUAR`
- **ZBrain Trade CSR** · ZBrain_Trade_CSR · `00GdM00000Pgw6vUAB`
- **ZBrain SOM CSR** · ZBrain_SOM_CSR · `00GdM00000PgwyUAB`
- **ZBrain CTA (Contracts & Agreements)** · ZBrain_CTA_Scope · `00GdM00000PgxtUAR`
- **ZBrain Sales Order Owner / POB** · ZBrain_POB_Owner · `00GdM00000Pgxt2UAB`

Each is a Group of Type='Queue' with a QueueSObject binding for the Case object. Idempotent — re-running Provision adopts existing queues by DeveloperName rather than creating duplicates. Sync from Salesforce refreshes the queue_id + label on every owner_mapping KB row.